

Chapter 6

AI for Physical, Cognitive, and Developmental Challenges



Junjie Wang and Melissa D. Begg

1 Introduction

The World Health Organization (WHO) has reported that approximately 1.3 billion individuals worldwide—equating to about 16% of the global population, or 1 in 6 persons—experience significant disabilities. Empirical evidence suggests that some individuals with disabilities face a reduction in life expectancy of up to 20 years compared to their non-disabled counterparts. Furthermore, they are at a twofold increased risk for developing chronic health conditions, including depression, asthma, diabetes, stroke, obesity, and poor oral health. Substantial systemic and logistical barriers compound these health disparities; for instance, they experience difficulties in accessing transportation—both in terms of availability and affordability—at a rate 15 times greater than those without disabilities. These pervasive inequities are largely attributable to structural factors such as stigma, discrimination, poverty, and exclusion from educational and employment opportunities, as well as inherent barriers within the healthcare system itself (WHO, 2023).

Artificial intelligence (AI) is a transformative technology with significant implications across diverse sectors, particularly healthcare and business operations (Dhopte & Bagde, 2023). AI encompasses technologies that enable computers and machines to simulate human cognitive functions, such as learning, comprehension, problem-solving, decision-making, creativity, and autonomy (Stryker & Kavlakoglu, 2024). Its rapid advancements offer transformative potential to enhance support systems for persons with disabilities. In healthcare, AI-driven innovations—such as diagnostic tools and predictive analytics—enable the early detection of chronic

J. Wang (✉)

School of Sports and Health Promotion, Dalian University of Technology, Dalian, China

e-mail: wangjunjie@dlut.edu.cn

M. D. Begg

School of Social Work, Columbia University, New York, NY, USA

e-mail: mdb3@columbia.edu

conditions like depression, diabetes, and cardiovascular disease. These innovations facilitate timely, personalized interventions that help reduce premature mortality (Davenport & Kalakota, 2019) and promote systemic changes in diagnostic and clinical management processes (Thacharodi et al., 2024). In transportation, AI applications can optimize route planning and service scheduling, thereby improving the accessibility and affordability of mobility solutions for individuals facing logistical challenges (Olawade et al., 2025). Furthermore, AI-based assistive technologies—including speech recognition software, smart home devices, and adaptive interfaces—can improve independent living and daily navigation, fostering greater participation in education and employment and ultimately contributing to enhanced social inclusion (Boscardin et al., 2024).

Addressing the multifaceted challenges posed by AI in the physical, cognitive, and developmental domains for individuals with disabilities, this chapter focuses on key considerations. Specifically, it examines adaptive devices, communication tools, and rehabilitation technologies. Furthermore, it explores the crucial aspects of accessibility and user-centered design. Most importantly, it interrogates the ethical considerations of reliance on AI and the importance of incorporating the perspectives of those with lived experience. Central to all of this is the notion that AI can be deployed in ethical and valuable ways, so long as we are mindful of who is at the table and how biases can be mitigated—making the role of social workers critical in its development and impact. In the words of Tim Wu (2025), “Instead of asking which future is coming, we should be asking which future we want” (para. 4). Now is the time to think about how to craft that future.

2 AI in Addressing Physical Challenges

Physical challenges among individuals with disabilities encompass difficulties in performing everyday tasks due to impairments in movement, strength, coordination, and overall physical function. Recent advances in AI have led to significant improvements in the detection, diagnosis, and management of these challenges. AI-based techniques are increasingly being developed and implemented to address issues such as muscle degeneration, impaired mobility, and the need for enhanced rehabilitation strategies across diverse patient populations. These innovations facilitate earlier diagnosis and support personalized treatment planning, ultimately having the potential to improve patient outcomes and quality of life.

2.1 Advancements in AI-Driven Adaptive Devices

AI-powered assistive technologies integrate advanced AI—encompassing machine learning (ML), computer vision, and natural language processing—with conventional support devices to enhance functionality, personalization, and adaptability for

individuals with disabilities. By processing real-time data from sensors, cameras, and other sources, these systems detect user needs, predict behaviors, and provide tailored assistance across domains such as mobility, communication, and daily living tasks.

Recent AI advancements have significantly expanded the range of assistive technology applications. These breakthroughs span various categories, from wearable devices that augment sensory perception to sophisticated communication tools and the integration of AI into everyday smart environments (Table 6.1). Significant scientific interest has emerged in the development of AI-driven robotic exoskeletons for upper-limb applications. For example, an AI-driven universal exoskeleton system has been designed to dynamically switch assistance types across walking modes, modulate support based on ground slope, and provide timely intervention during specific gait phases. Such innovations have the potential to markedly improve the quality of life for individuals with physical disabilities by promoting independence and social inclusion (Lee et al., 2024). Similarly, AI-driven knee exoskeletons that utilize integrated sensors to gather pressure, movement, and force data have

Table 6.1 AI-powered assistive technologies

Technology name	Disability addressed	Key AI features
Ray-Ban Meta glasses (Bikkani, 2025)	Visual impairment	Real-time audio descriptions of environments
Orion (Bikkani, 2025)	Visual impairment	Multitasking assistance, holographic displays
NeuroVision Pro (Carr, 2024)	Blindness/visual impairment	“Digital sight” via neural signals from cameras, sensors, and AI
BionicM’s Bio Leg (Bikkani, 2025)	Mobility impairment	Adapts to user movements, learns gait patterns
ExoWalk (Carr, 2024)	Mobility disabilities	Lighter, more sensitive exoskeleton with AI-based motion prediction
Auracast (Bikkani, 2025)	Hearing impairment	Broadcasts audio to multiple devices simultaneously without individual pairing
Otter.ai (Bikkani, 2025)	Hearing/language barriers	Real-time multilingual translation
Samsung’s Bulging TV (Bikkani, 2025)	Sensory disabilities	Tactile 3D screens with haptic feedback
SoundSphere (Carr, 2024)	Hearing impairment	Personalized 360-degree soundscape with AI filtering
Accessible Google Maps (Bikkani, 2025)	Mobility/visual impairment	Wheelchair-friendly routes, better live view, disabled-owned business labels
GPT-4 + Be my Eyes (Bikkani, 2025)	Visual impairment	AI-powered virtual volunteer assistance
NanoGlove (Carr, 2024)	Limited hand mobility	AI-based adaptability tailors
Alinker (Carr, 2024)	Mobility impairment	Non-motorized walking bike for social inclusion and physical activity
HoverChair (Carr, 2024)	Mobility impairment	Omnidirectional movement and standing with AI control

been developed to deliver adaptive gait assistance (Puyuelo-Quintana et al., 2020). Another system leverages electromyography signals to enable human-robot cooperative movement, providing assistance tailored to user needs (Lyu et al., 2019).

Furthermore, AI-powered assistive technologies extend beyond mobility support. For instance, an eye-tracking control system for assistive robotics has demonstrated a 100% task-completion rate among individuals with motor impairments, thereby enhancing the efficacy of wheelchair and robotic manipulator control while promoting user engagement and motivation (Sunny et al., 2021). In the realm of in-home care, AI systems that analyze real-time sensor data have shown promise in predicting falls and other adverse health events among seniors, thereby facilitating timely interventions and improving overall care (Momin et al., 2022). Integrating AI with various agents—including electronics, robotics, and software—continues to revolutionize assistive technologies, enabling people with disabilities to achieve greater independence in daily activities, access community and healthcare services more effectively, and enhance productivity in educational and employment settings (Pancholi et al., 2024).

AI-driven adaptive devices can significantly enhance the quality of life for individuals with disabilities by offering personalized assistance, real-time environmental adaptation, and improved accessibility. These technologies, such as AI-powered prosthetics, speech-to-text systems, and smart navigation tools, foster greater independence and social inclusion.

2.2 AI Application in Physical Therapy

The transformative potential of AI, and specifically ML, for clinical practice is increasingly evident. In neurorehabilitation, AI is revolutionizing treatment paradigms by offering personalized, data-driven interventions that enhance recovery outcomes in patients with neurological disorders such as stroke, spinal cord injury, and Parkinson's disease (PD) (Morelli, 2024). By analyzing intricate patterns within diverse datasets, AI algorithms can identify subtle indicators that inform both diagnosis and treatment planning (Calderone et al., 2024). A notable development in this area is the synergistic integration of wearable technologies with AI. These devices provide continuous data streams by monitoring patient movements and physiological responses outside traditional clinical environments. These data, when incorporated into ML models, enable real-time assessment of intervention efficacy and facilitate timely adjustments to treatment protocols (Swarnakar & Yadav, 2023). This iterative process of monitoring and refinement is particularly valuable in therapies aimed at regaining muscle strength and improving balance, where incremental gains can significantly enhance a patient's quality of life (Sardari et al., 2023).

Furthermore, AI-powered decision support systems are being developed to assist clinicians in formulating optimal intervention strategies. By analyzing historical data from large patient cohorts—including imaging studies, wearable sensor outputs, and patient demographics—these systems forecast which individuals are most

likely to benefit from specific therapeutic approaches. This holistic assessment not only augments clinical decision-making but also promotes more efficient allocation of healthcare resources, potentially reducing the overall disease burden (Johnson et al., 2021; Morelli, 2024). Despite these advancements, it remains imperative that AI and ML tools complement rather than replace human expertise in clinical settings. The integration of these technologies into routine practice should support healthcare providers in delivering more effective, personalized care while ensuring that the nuanced judgment and empathy inherent to human practitioners remain central to patient management (Fiorente et al., 2024). The development and deployment of AI technologies must ensure that care decisions and care delivery continue to center on the individual patient.

Future directions in AI for addressing physical challenges are likely to focus on enhancing the interoperability of wearable devices, improving data security and privacy, and refining algorithmic models for even more precise predictions. As AI continues to evolve, its integration with Internet of Things (IoT) and personalized rehabilitation strategies is expected to lead to more adaptive, responsive, and effective treatment modalities. This integration will pave the way for a new era in physical healthcare, where continuous, data-driven insights empower clinicians and patients alike to achieve better health outcomes through tailored, proactive interventions (Sumner et al., 2023).

3 AI in Addressing Cognitive Challenges

Cognitive challenges refer to impediments to efficient cognitive processing, including deficits in attention, memory, problem-solving, and decision-making. These challenges manifest in educational, professional, and everyday contexts, affecting performance and overall satisfaction. Recognizing and addressing these limitations is essential for optimizing personal development, productivity, and learning strategies. Given that individuals experience cognitive challenges to varying degrees, tailored interventions are necessary to enhance cognitive efficacy (Anderson, 2024).

Recent advancements in AI have significantly contributed to the detection, diagnosis, and management of cognitive impairments. AI-based techniques are increasingly applied to address a spectrum of challenges—from autism spectrum disorder (ASD) and PD to rehabilitation in diverse patient populations. These innovations not only improve healthcare delivery but also enable personalized learning experiences by adapting to individual needs and pacing. By analyzing performance data, AI systems can offer customized content and feedback, thereby enhancing cognitive skills and promoting better knowledge retention (Gupta et al., 2023), which is essential to patient-centered care.

3.1 *AI in Detection and Diagnosis*

Recent advances in AI have demonstrated significant promise in the early detection and diagnosis of various cognitive disorders, including PD, ASD, Alzheimer's disease (AD), Attention-Deficit Hyperactivity Disorder (ADHD), addiction, and sleep disorders (Bazoukis et al., 2023; Namasse et al., 2025; Perju-Dumbrava et al., 2022; Sun et al., 2023; Suva & Bhatia, 2024; Winchester et al., 2023).

For example, in PD detection, AI models have achieved an area under the curve of 0.90 on held-out test sets and 0.85 on external test sets. These models also reliably estimate disease severity and progression using established clinical benchmarks such as the Movement Disorder Society Unified Parkinson's Disease Rating Scale (Yang et al., 2022). Data mining of prodromal markers—ranging from cognitive disturbances to cerebrospinal fluid measurements—has yielded diagnostic accuracies as high as 96%, with sensitivities and specificities reaching 97% and 95%, respectively (Prashanth et al., 2016). Moreover, ML models capturing non-linear and overlapping disease trajectories have provided a comprehensive view of early PD progression (Severson et al., 2021).

In the realm of neurodevelopmental disorders, the standard diagnostic process for ASD has traditionally relied on clinician-administered behavioral observations, a method often hampered by subjectivity and time constraints. In contrast, MRI-based AI techniques have demonstrated improved objectivity, speed, and accuracy in early detection of ASD (Helmy et al., 2023). Similarly, early indicators of AD, such as speech and language alterations, have been effectively analyzed using AI-driven natural language processing. This non-invasive and cost-effective approach facilitates the timely initiation of management strategies and interventions (Jahan et al., 2024). This can lead to substantial patient benefits.

Additional ML models further underscore AI's diagnostic potential. One model, integrating support vector machines (SVMs) with principal component analysis (PCA), was developed to predict cognitive decline in PD using clinical data, biomarkers, and neuropsychological tests. This model achieved 92.3% accuracy with 32 parameters, which improved to 100% after feature selection reduced the parameter set to 13 (Chen et al., 2022a). Another model demonstrated high logistic classification performance, achieving 91.67% accuracy in classifying mild cognitive impairment subtypes using gait data (Chen et al., 2020).

Overall, integrating AI into early detection and diagnosis not only improves the precision of traditional assessments but also accelerates the diagnostic process, paving the way for earlier and more effective interventions.

3.2 *AI in Cognitive Rehabilitation Therapy*

In March 2020, the National Institute on Aging (NIA) launched the Artificial Intelligence and Technology Collaboratory for Aging Research to accelerate the adoption of AI and digital technologies. This initiative specifically targets improvements in health and quality of life for older adults, particularly those affected by dementia and AD (Abadir et al., 2024).

AI has emerged as a transformative tool in cognitive rehabilitation by enhancing the adaptability of therapeutic interventions. ML algorithms now adjust exercise difficulty in real time according to individual patient performance, ensuring that challenges remain optimally calibrated to patient-identified needs and preferences. In addition, AI's predictive capabilities facilitate early identification of setbacks or plateaus in recovery, enabling timely, proactive modifications to treatment plans (Bhattarai et al., 2023; Campagnini et al., 2022; Chen et al., 2023).

Cognitive Rehabilitation Therapy (CRT) is a non-pharmacological intervention that employs both restorative and compensatory strategies to enhance functional cognition in daily living activities (Khalid et al., 2024). The integration of AI into CRT has led to the development of advanced AI-powered robots designed to enhance essential cognitive functions, such as memory retention, focus, attention span, and executive skills. Notably, AI-enabled social robots like PARO have demonstrated tangible benefits by offering continuous support and personalized cognitive assistance across various settings, including healthcare and social care (Lu et al., 2021). Furthermore, AI-driven socially assistive robots—whether designed in animal, robotic, or humanoid forms—have shown promising results in improving cognitive function and can serve as a digital and social prescription program, providing 24-h engagement for older adults (Lee et al., 2022).

In addition, virtual reality (VR) systems powered by AI create immersive environments that simulate real-life scenarios, thereby stimulating neuroplasticity and enhancing cognitive recovery. The synergistic application of VR and Brain–Computer Interface (BCI) enables targeted neuromodulation of specific brain areas, potentially eliciting neurochemical shifts and influencing cognitive functions, including memory, perception, and motor skills (Drigas & Sideraki, 2024). Integrating AI and VR into chronic pain cognitive therapy shows promise for enhanced management and improved patient outcomes (Mazzolenis et al., 2024). Similarly, AI-driven VR platforms personalize cognitive rehabilitation, monitor progress, and show promising results for dementia (Velagaleti, 2025). These advancements exemplify AI's enormous potential to bridge the gap between clinical expertise and innovative technology, ultimately leading to more effective and efficient cognitive rehabilitation strategies.

Despite these advancements, challenges remain, and the greatest promise of AI in addressing cognitive challenges may lie in personalized cognitive augmentation. AI, converging with cognitive science, will evolve beyond pattern recognition to offer real-time, non-disruptive interventions for cognitive biases—which are systematic and predictable deviations from rational judgment or decision-making

(Blanco, 2017)—and limitations. Personalized platforms will tailor support for executive functions, benefiting neurodivergent and aging populations by preserving cognitive autonomy. Overall, the strategic incorporation of AI in CRT holds significant promise for revolutionizing patient care and advancing the field of neurorehabilitation (Gupta, 2024).

4 Accessibility, User-Centered Design, and the Digital Divide

4.1 Accessibility Challenges in AI

The integration of AI into various facets of life necessitates a critical examination of its accessibility for individuals with disabilities. Digital accessibility, as defined by established initiatives, ensures equitable access to digital content and services for everyone, regardless of ability. This relies on the POUR principles: perceivable, operable, understandable, and robust design (Chemnad & Othman, 2024). While AI holds immense potential to enhance accessibility and improve the quality of life for people with disabilities, its current trajectory presents several challenges that demand scholarly attention (Liyana et al., 2019). The increasing reliance on AI in decision-making processes further underscores the importance of ensuring these systems are comprehensible and usable by all, including those with disabilities. Failure to prioritize accessibility in the development and deployment of AI risks creating new forms of exclusion and discrimination, underscoring the ethical imperative to address these challenges proactively (Fjeld & Brynn, 2024).

A significant logistical hurdle in creating accessible AI is the availability of diverse, representative datasets for training models. AI algorithms learn from patterns in data, and if disability is underrepresented or inaccurately categorized, the resulting models will not effectively serve individuals with disabilities (Lay-Flurrie, 2025). Similarly, speech recognition systems may perform poorly for individuals with speech impairments due to the lack of diverse speech samples in their training data (Goudarzi & Moya-Galé, 2021). Many readily available datasets tend to reflect the most common data, inadvertently excluding individuals with disabilities who have unique accessibility requirements. This lack of representative labeling can also lead to misclassifications in crucial accessibility features like alternative text for images, impacting nonbinary individuals and those with disabilities (Bennett & Trewin, 2023). The scarcity of comprehensive disability data, termed a “disability data desert,” significantly limits AI utility and can inadvertently perpetuate harmful stereotypes and amplify existing biases.

Furthermore, high costs and market failures limit access to advanced AI-assistive technologies. The necessity for sophisticated hardware and extensive computational resources creates an economic barrier, particularly for low-income regions and underserved populations. Complex interfaces requiring high technical literacy further restrict adoption (Botelho, 2021). Infrastructural deficits, encompassing limited

network connectivity and intermittent power supply, pose significant impediments (Deo & Anjankar, 2023). These limitations create substantial challenges for the deployment and effective operation of AI in various contexts.

4.2 User-Centered Design Principles in AI

User-centered design (UCD) emphasizes placing the user at the heart of the development process. This approach involves iterative cycles of prototyping, testing, and refinement to ensure that solutions are tailored to real-world needs (Luna et al., 2015). In the context of AI, UCD requires a rigorous analysis of user behaviors, contextual challenges, and environmental factors (McDermott et al., 2023). It promotes empathy, usability, and accessibility by focusing on the users' experiences and limitations. Methodologies such as contextual inquiry, heuristic evaluation, and participatory design are essential. These techniques help uncover hidden requirements and contextual nuances that may be overlooked in traditional design approaches, ensuring that the final product is both intuitive and accessible (El et al., 2024) and tailored and responsive to the needs and preferences of those affected.

Direct engagement with disabled users is paramount to creating inclusive AI systems. Involving these users from the conceptual stage through to the final evaluation helps identify specific challenges that may not be evident to designers without lived experiences of disability. This participatory design process not only enriches the product's development but also empowers users by validating their input (Jung et al., 2025). For example, incorporating user feedback on interface adjustments, navigation methods, and accessibility features enables designers to address diverse needs effectively. From the very start of the design process, regular workshops, focus groups, and usability testing sessions provide platforms for dialogue and collaborative problem-identification and problem-solving. Moreover, such involvement ensures that design decisions are informed by practical experiences rather than theoretical assumptions, thereby increasing the likelihood of adoption and long-term user satisfaction.

The effectiveness of AI-based solutions depends heavily on their ease of use, flexibility, and adaptability. Ease of use is achieved through intuitive interfaces that reduce cognitive load and learning curves. This is particularly important for users with varying levels of digital literacy or who require assistive technologies. Flexibility and adaptability, on the other hand, refer to the system's ability to accommodate a wide range of user needs and contexts. This involves creating modular systems that can be customized based on individual requirements, such as adjustable display settings, voice-controlled commands, or alternative input modalities. Additionally, adaptive algorithms that learn from user interactions can further refine system performance over time. These design considerations not only enhance usability but also ensure that AI solutions remain relevant in rapidly changing technological landscapes (Prince et al., 2024).

4.3 *The Digital Divide in AI*

The pervasive integration of digital technologies has fundamentally reshaped societal structures, yet this transformation has not been uniformly experienced across all demographics and regions. The term “digital divide” traditionally describes the unequal access to digital technologies, encompassing disparities in Internet connectivity, device availability, infrastructure, skills, and affordability, which consequently result in unequal opportunities for information access and digital participation (Lythreathis et al., 2022; Wong et al., 2025). As AI rapidly advances and its applications become increasingly widespread, a new dimension of this disparity has emerged: the “AI divide” (i.e., AI-related access, capability, or outcome divide among users) (Carter et al., 2020). This evolving divide signifies unequal access to, benefits from, and opportunities presented by AI across regions, communities, and socioeconomic strata (Rajpurkar et al., 2022).

The AI divide manifests across several interconnected facets, creating a complex web of disparities. One significant aspect is the unequal access to AI technologies. The development and deployment of AI are concentrated primarily in wealthy nations and major technology companies. Developed countries typically enjoy robust technological infrastructure, substantial investment in research and development, and higher digital literacy rates, which collectively facilitate the integration of AI in assistive devices. In contrast, developing countries often face outdated or inadequate digital infrastructure, high costs of modern technology, and limited access to technical training. This concentration exacerbates the global imbalance, creating a digital hierarchy that marginalizes underdeveloped areas (Ahmad et al., 2024).

Another critical facet of the AI digital divide lies in the disparities in AI education and skills (Nagi et al., 2023). A substantial gap exists between the growing demand for AI expertise and the current supply. This gap exhibits demographic imbalances, with studies indicating lower AI training and proficiency among women and older adults than among men and younger adults, respectively (Randstad, 2025). Educational barriers in underserved regions further exacerbate this issue by limiting access to foundational digital literacy and AI-related skills. Psychological factors, such as technology resistance (which may be partly rooted in mistrust stemming from persistent societal inequities), also hinder engagement, particularly among those with limited prior exposure.

The third critical dimension of the AI divide lies in the unequal availability of relevant data for AI development. Fundamentally data-driven, effective AI systems rely on large, diverse, and high-quality datasets for training and improvement (Tripathi et al., 2023). Consequently, limited access to such data creates a significant disadvantage. A key concern is the pervasive issue of bias within training data, which can lead to discriminatory outcomes for marginalized groups. This bias manifests through geographical limitations, underrepresentation of racial and ethnic minorities and medically underserved populations, and other vulnerable groups, and linguistic and cultural barriers—ultimately hindering the development of fair

and inclusive AI applications, especially in critical domains like healthcare (Dankwa-Mullan, 2024).

5 Ethical Considerations in Inclusive AI Development

The development of AI prediction algorithms involves several stages, each of which may introduce bias (Nazer et al., 2023). Algorithms do not create biases themselves but may perpetuate (or even amplify) societal inequities and cultural prejudices. Bias can enter at different stages of development and deployment, including data sets, algorithms, and systems (Chen, 2023). The reasons may vary, including a lack of access to data for target populations, unconscious and conscious bias within the developing team, and organizational structure and practices. As a result, algorithms may provide inaccurate predictions and outputs for certain subsets of the population or discriminate against particular groups (Pasipamire & Muroyiwa, 2024). AI algorithm development stages and potential bias sources are outlined for ethical consideration.

5.1 *Formulating the Research Problem*

This initial stage focuses on defining the research question or problem. Potential biases introduced here include:

- **Racial/Ethnic Bias:** Societal prejudices rooted in race and ethnicity can introduce framing bias. This influences the conceptualization of problems and potential solutions (Skinner-Dorkenoo et al., 2023).
- **Gender Bias:** Bias arising from societal stereotypes and inequalities related to gender that can influence the research question (Kotek et al., 2023).
- **Age Bias:** Bias resulting from age-related stereotypes or the exclusion of specific age groups within the problem definition, leading to skewed outcomes (Reio, 2011).
- **Disability Bias:** Bias that occurs when the problem formulation does not adequately consider or address the needs and perspectives of individuals with disabilities (Antonopoulos et al., 2023).
- **Linguistic Bias:** Linguistic bias is a systematic preference for specific language forms that leads to inequitable treatment or evaluation of individuals or groups based on their language use, potentially through the language used to define the problem (Beukeboom & Burgers, 2017).
- **Global Misrepresentation Bias:** Bias arises from the underrepresentation or inaccurate portrayal of specific populations or regions within a research problem (Shahbazi et al., 2023).

5.2 *Data Collection*

This stage involves gathering the data needed to train and evaluate the machine learning model. Biases that can arise during data collection are:

- **Sampling Bias:** Bias introduced when the collected data is not representative of the overall population of interest (Chen et al., 2022b).
- **Measurement Bias:** Bias resulting from systematic errors or inconsistencies in how data is measured or recorded (Hernán & Cole, 2009).
- **Exclusion Bias:** Bias caused by the intentional or unintentional exclusion of certain groups or data points from the dataset (Favier et al., 2023).
- **Label Bias:** Bias present in the labels assigned to the data, which can reflect existing societal biases or flawed annotation processes (Favier et al., 2023).
- **Social Bias:** Bias embedded in the data that reflects and potentially amplifies societal stereotypes and prejudices (Shin et al., 2024).

5.3 *Data Pre-Processing*

This stage involves cleaning, transforming, and preparing the collected data for model training. Potential biases introduced during pre-processing include:

- **Aggregation Bias:** Bias that occurs when data is combined in a way that masks or distorts essential differences between subgroups (Voss, 2005).
- **Feature Selection Bias:** Bias introduced when the features chosen to train the model are more relevant or accurate for some groups than others (Singhi & Liu, 2006).
- **Outlier Bias:** Bias caused by the disproportionate influence of outliers in the dataset, which might not be representative of all groups (Ding et al., 2024).
- **Confounding Bias:** Bias arising when a hidden or unobserved variable is correlated with both the input features and the outcome, leading to misleading associations (Schuster et al., 2021).

5.4 *Model Development and Validation*

This stage involves selecting a model architecture, training it on the pre-processed data, and evaluating its performance. Biases at this stage can include:

- **Training Dataset Bias:** Bias present in the data used to train the model, leading it to learn and perpetuate existing inequalities (Mac Namee et al., 2002).
- **Test Dataset Bias:** Bias in the data used to evaluate the model, potentially leading to an inaccurate assessment of its performance across different groups (Hou et al., 2024).

- **Algorithmic Bias:** Bias inherent in the design or assumptions of the machine learning algorithm itself, which might favor certain outcomes or groups (Hooker, 2021).
- **Confirmation Bias:** Bias where the model development or evaluation process is influenced by pre-existing beliefs or expectations, leading to the overlooking of potential biases (Taehyun & Sangyeon, 2023).
- **Validation Bias:** Bias introduced during model validation, potentially leading to an overly optimistic estimate of the model's performance on unseen data (Nazer et al., 2023).

5.5 Model Implementation

This final stage involves deploying the trained model for real-world use. Biases that can emerge during implementation include (Nazer et al., 2023):

- **Concept Drift:** Bias that occurs when the statistical properties of the data processed by the model change over time, leading to degraded performance or unfair outcomes.
- **Covariate Shift:** Bias that arises when the distribution of input features in the deployment environment differs significantly from that in the training data.

6 Conclusion

AI has demonstrated its potential to significantly enhance the quality of life for people with disabilities through a variety of applications and technologies. AI-powered assistive technologies offer the promise of more personalized, adaptive, and effective support tailored to each individual's unique circumstances. The integration of AI into assistive technologies is poised to revolutionize the lives of individuals with disabilities, making accessibility more effortless and breaking down barriers that once seemed insurmountable. The focus of innovation is on creating tools that are not only technologically advanced but also simple, accessible, and versatile, catering to a wide range of needs and preferences. The concept of assistive augmentation, which envisions technology as an integrated and seamless extension of human ability, represents a key direction for future development in this field (Tan et al., 2025). However, realizing this potential hinges on careful navigation of ethical considerations, incorporating the perspectives of those with lived experience, and a steadfast commitment to ensuring accessibility for all. Continued collaboration between researchers, developers, policymakers, and, most importantly, individuals with disabilities is crucial for realizing the full transformative potential of AI-based assistive technologies. The overarching trend is toward the development of more integrated, personalized, and intelligent assistive solutions that will empower individuals with disabilities to lead more independent, fulfilling, and engaged lives.

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